

## Introduction

Diversity metrics summarise microbial community composition. **Alpha diversity** measures richness and evenness within a sample; **beta diversity** measures compositional dissimilarity between samples. Both are essential for comparing communities and detecting treatment, disease, or niche effects.

## Prerequisites

ASV or OTU count table; phylogenetic tree (for UniFrac); sample metadata.

## Theory

**Alpha diversity:** - **Richness** (# ASVs observed). - **Shannon index**  $H = -\sum p_i \log p_i$  – evenness-weighted.  
- **Simpson**  $1 - \sum p_i^2$  – inverse concentration. - **Faith's PD** – phylogenetic tree branch length sum.

**Beta diversity:** - **Bray-Curtis** – abundance-weighted Manhattan dissimilarity. - **Jaccard** – presence/absence. - **UniFrac (weighted / unweighted)** – phylogeny-aware.

Permutational MANOVA (`adonis2`) tests whether group membership explains beta diversity.

## Assumptions

Counts are non-negative; rarefaction or equivalent normalisation is applied before alpha diversity if depths vary widely; phylogenetic tree matches ASV labels.

## R Implementation

```
library(vegan)

set.seed(2026)
n_samples <- 20; n_asv <- 50
counts <- matrix(rpois(n_asv * n_samples, lambda = 10),
                 n_samples, n_asv) # samples x ASVs

# Inject group differences
group <- factor(rep(c("ctrl", "case"), each = 10))
counts[group == "case", 1:15] <- counts[group == "case", 1:15] + 20

shannon <- diversity(counts, index = "shannon")
simpson <- diversity(counts, index = "simpson")
richness <- rowSums(counts > 0)

by(shannon, group, median)

bray <- vegdist(counts, method = "bray")
adonis_res <- adonis2(bray ~ group, permutations = 999)
adonis_res
```

## Output & Results

Shannon / Simpson / richness per sample (group medians printed), and a PERMANOVA on Bray-Curtis showing whether group explains dissimilarity.

## Interpretation

“Case samples had lower Shannon diversity (median 2.1 vs 2.9, Wilcoxon  $p = 0.004$ ) and PERMANOVA on Bray-Curtis revealed significant community-level differences ( $R^2 = 0.18$ ,  $p = 0.001$ ).”

## Practical Tips

- Rarefy to equal depth before alpha diversity to avoid depth-driven confounding; alternatively use rarefaction curves to confirm coverage.
- `phyloseq::estimate_richness` computes many alpha metrics in one call.
- Weighted UniFrac captures abundance-weighted phylogenetic distance; unweighted is presence-only.
- PERMANOVA is sensitive to dispersion heterogeneity; run `betadisper` to confirm equal dispersion before interpreting.
- PCoA visualisation on the beta dissimilarity matrix complements PERMANOVA.

## For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

## See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat